ONLINE EXAMINATION SYSTEM

21CSC205P Database Management Systems
A MINI PROJECT REPORT
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BONAFIDE CERTIFICATE

Certified that Project report titled

"ONLINE EXAMINATION SYSTEM"

is the bonafide work of

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who carried out the **21CSC205P Database Management Systems** mini project work under my supervision.

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ABSTRACT

The Online Examination System is a simple database-based software used to conduct exams through computers instead of paper. It allows students to register,log in, attend exams, answer multiple-choice questions, and receive their results automatically. The system stores all details about students, exams, questions, attempts, and results safely in a database, which makes the process organized and secure. Teachers or administrators can create exams, add questions, and manage student records easily without paperwork. Automatic evaluation reduces human errors and saves time in checking answer sheets and preparing results.

This project is designed using basic database concepts such as ER diagram, relational schema, tables, primary keys, and foreign keys to connect all the data correctly. SQL commands are used to create the database, insert data, and manage operations. The relationships between student, exam, question, attempt, and result are clearly defined to avoid data duplication and maintain accuracy. The database structure helps in quick retrieval of information whenever required.

The system is user-friendly and can be used in schools, colleges, training centers, and online certification platforms. It supports smooth exam management, proper data storage, and easy result processing. Since everything is stored digitally, records can be maintained for a long time without loss of data.

The Online Examination System also improves transparency in the examination process. Students can view their performance clearly, and administrators can track exam activities and student attempts easily. This makes monitoring and managing exams more effective.

In addition, the system reduces the use of paper and supports an eco-friendly approach. It minimizes manual work, reduces the chances of data loss, and ensures that exam data is always available when needed.

Overall, the Online Examination System provides a fast, reliable, secure, and efficient way to conduct examinations digitally while showing the practical implementation of Database Management System concepts in a real-world application.

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CHAPTER 1

PROBLEM UNDERSTANDING, IDENTIFICATION OF ENTITY AND RELATIONSHIPS, CONSTRUCTION OF DB USING ER MODEL FOR THE "ONLINE EXAMINATION SYSTEM"

1.1 Introduction

An Online Examination System is a software application used to conduct exams through computers or the internet. Students can log in, attend exams, answer questions, and receive results automatically. This system removes the need for paper-based exams and makes the process faster, easier, and more accurate.

1.2 Motivation

Traditional exams require paper, manual correction, and a lot of time. There can be mistakes in evaluation and result processing. The Online Examination System solves these problems by:

- Conducting exams online
- Automatic evaluation of answers
- Instant result generation
- Secure storage of student and exam data

1.3 Scope

This system can be used in:

- Schools and colleges
- Training institutes
- Competitive exams
- Online certification tests

It manages students, exams, questions, attempts, and results.

1.4 Problem Statement

Manual examination systems are slow, require paperwork, and are prone to errors. There is a need for a computerized system that can manage exams, students, questions, and results efficiently.

The Online Examination System provides a database-driven solution to conduct and manage exams online.

1.5 Project Requirements

The system requires:

- Student registration and login
- Exam creation
- Question management
- Students attempting exams
- Automatic score calculation
- Result generation with grade and status

1.6 Identification of Entity and Relationships

Entity	**Description**
STUDENT	Stores student details
EXAM	Stores exam details
QUESTION	Stores questions of exams
ATTEMPTS	Stores student exam attempt details
RESULT	Stores result after exam

Entities and Attributes

1.STUDENT

- student_id (PK)
- first_name
- last_name
- dob
- email_id
- password
- phone_no

2.EXAM

- exam_id
- exam_name
- date
- duration

3.QUESTION

- question_id (PK)
- exam_id (FK)
- question_text
- marks
- question_type

4.ATTEMPTS

- attempt_no (PK)
- student_id (FK)
- exam_id (FK)
- start_time
- end_time

5.RESULT

- Score
- result_id
- attempt_no
- grade
- Status

6. OPTIONS

- option_id (PK)
- question_id (FK)
- option_text
- is_correct

7. ANSWER

- answer_id (PK)
- attempt_id (FK)
- question_id (FK)
- option_id (FK)
- written_answer
- marks_obtained

Relationships

1. **Student → Attempts** (Makes)

- A student can make **many attempts.**
- Each attempt belongs to only **one student.**
- Relationship type: **1 : M**

2. **Exam → Attempts** (Has)

- One exam can have **many attempts by students.**
- Each attempt is for only **one exam**.
- Relationship type: **1 : M**

3. **Exam → Question** (Contains)

- One exam contains **many questions.**
- One question belongs to only **one exam.**
- Relationship type: **1 : M**

4. **Attempts → Result** (Generates)

- One attempt generates **one result.**
- Each result is based on only **one attempt.**
- Relationship type: **1 : 1**

5. Attempts → Answer (Records)

- One attempt records many answers.
- Each answer belongs to only one attempt.
- Relationship type: **1 : M**

6. Question → Answer (Is answered in)

- One question can appear in many answers (different students / attempts).
- Each answer is for only one question.
- Relationship type: **1 : M**

7. Question → Options (Has)

- One question can have many options.
- Each option belongs to only one question.
- Relationship type: **1 : M**

1.7 Construction of DB Using ER Model for the Online Examination System.

E-R Diagram

The ER diagram shows how students attend exams by making attempts.Each exam contains multiple questions. When a student attempts an exam, a result is generated based on the attempt. The ER model helps to clearly represent the relationships between student, exam, question, attempt, and result and is used to design the database tables without redundancy.

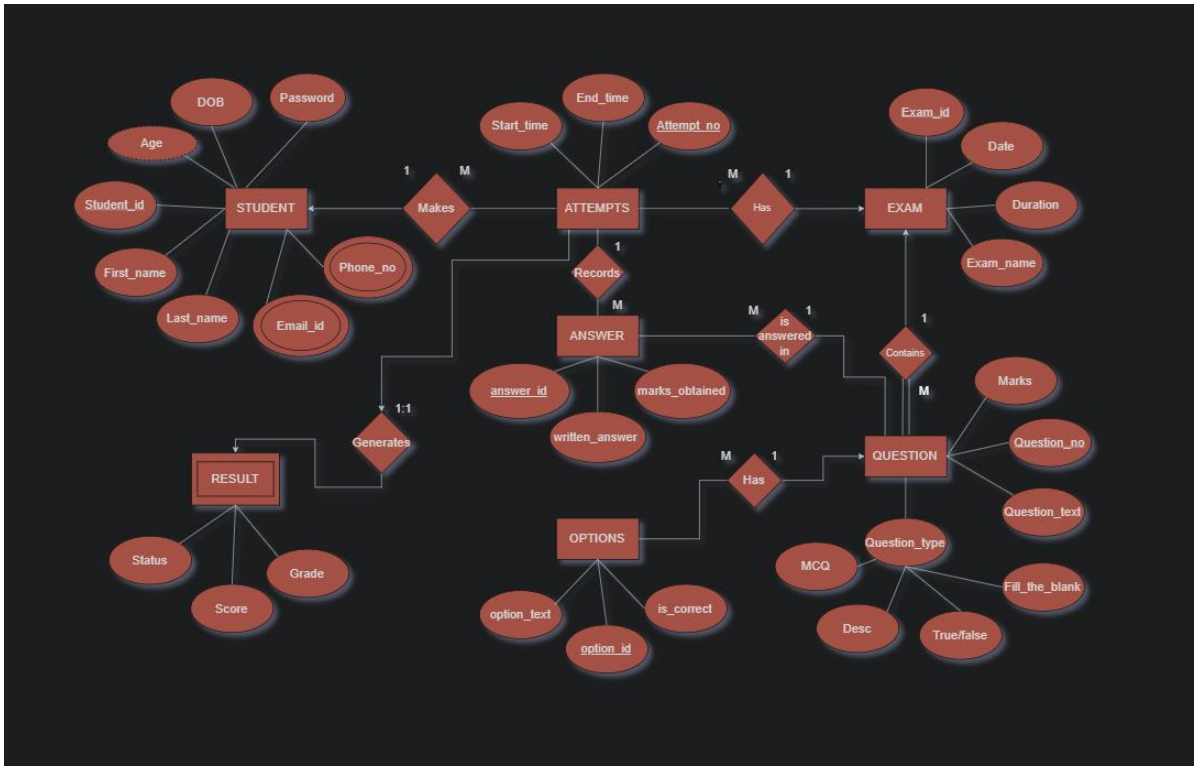

Figure 1.1: ER Diagram of Online Examination System.

Chapter 2

Design of Relational Schemas, Creation of Database and Tables for Online Examination System.

2.1 Relational Schema for Online Examination System.

The relational schema is derived from the ER diagram by converting each entity into a table and relationships into foreign keys.

Relational Schema:

- STUDENT(student_id PK, first_name, last_name, dob, email_id UNIQUE, password, phone_no)
- EXAM(exam_id PK, exam_name, date, duration)
- ATTEMPT(attempt_no PK, student_id FK, exam_id FK,start_time, end_time)
- QUESTION(question_id PK, exam_id FK, question_text, marks, question_type)
- OPTION(option_id PK, question_id FK, option_text, is_correct)
- ANSWER(answer_id PK, attempt_no FK, question_id FK, option_id FK, written_answer, marks_obtained)
- RESULT(attempt_no FK, score, grade, status)

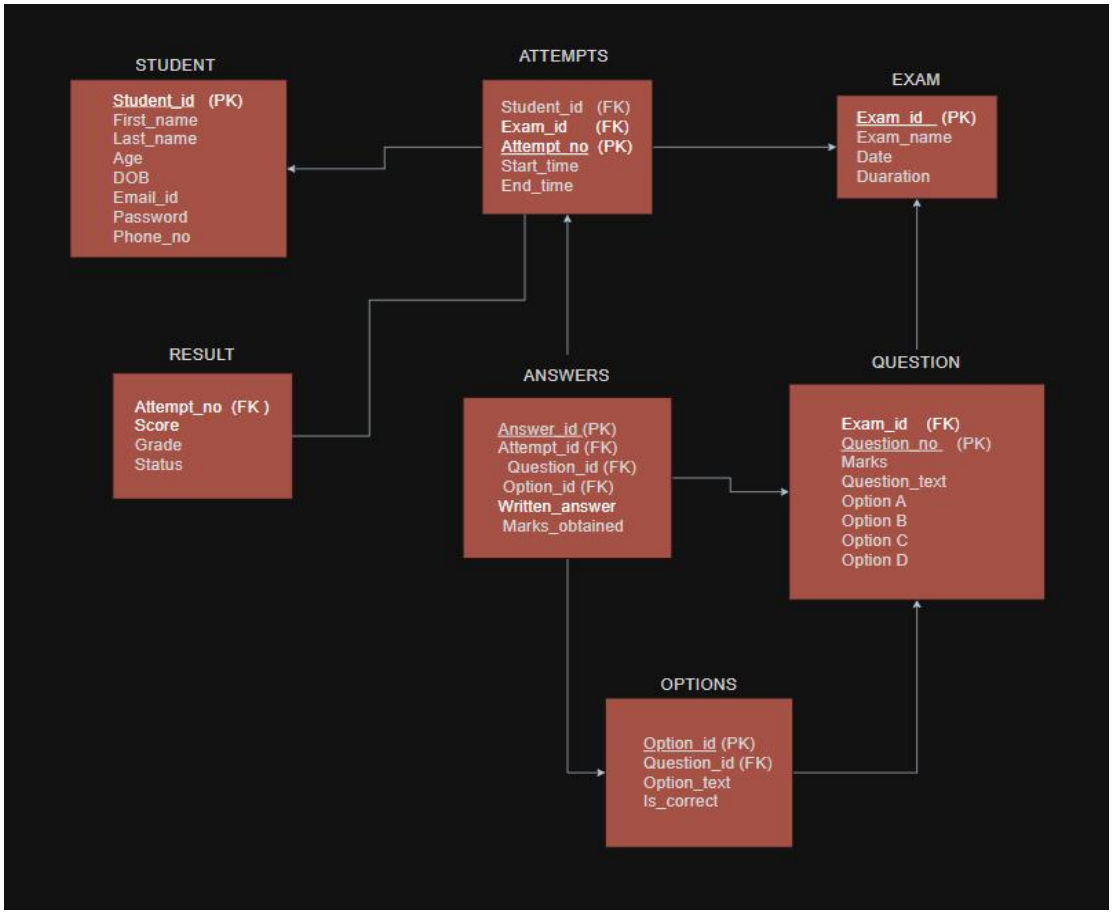

Figure 2.1: Relational Schema for Online Examination System.

The figure 2.1 depicts the relational schema for the Online Examination System.

The above schema shows how primary keys and foreign keys connect the tables to maintain proper relationships.

2.2 Description of Tables

STUDENT

Attribute	Data Type	Constraint
student_id	INT	Primary Key
first_name	VARCHAR(50)	Not Null
last_name	VARCHAR(50)	—
age	INT	—
email_id	VARCHAR(100)	Unique
password	VARCHAR(50)	Not Null
phone_no	VARCHAR(15)	—

EXAM

Attribute	Data Type	Constraint
exam_id	INT	Primary Key
exam_name	VARCHAR(100)	Not Null
date	DATE	—
duration	INT	—

QUESTION

Attribute	Data Type	Constraint
question_no	INT	Primary Key
exam_id	INT	Foreign Key
question_text	VARCHAR(255)	Not Null
marks	INT	—
optionA	VARCHAR(100)	—
optionB	VARCHAR(100)	—
optionC	VARCHAR(100)	—
optionD	VARCHAR(100)	—

ATTEMPTS

Attribute	Data Type	Constraint
attempt_no	INT	Primary Key
student_id	INT	Foreign Key
exam_id	INT	Foreign Key
start_time	TIME	—
end_time	TIME	—

OPTIONS

Attribute	Data Type	Constraint
option_id	INT	Primary Key
question_id	INT	Foreign Key
option_text	VARCHAR(100)	Not Null
is_correct	BOOLEAN	—

ANSWER

Attribute	Data Type	Constraint
answer_id	INT	Primary Key
attempt_id	INT	Foreign Key
question_id	INT	Foreign Key
option_id	INT	Foreign Key
written_answer	VARCHAR	—
marks_obtained	INT	—

RESULT

Attribute	Data Type	Constraint
result_id	INT	Primary Key
attempt_no	INT	Foreign Key
score	INT	—
grade	VARCHAR(5)	—

2.3 Creation of Database and Tables - DDL Commands

```
CREATE TABLE ATTEMPT(
  attempt_id INT PRIMARY KEY,
  student_id INT,
  exam_id INT,
  attempt_no INT,
  start_time TIME,
  end_time TIME,
  FOREIGN KEY (student_id) REFERENCES STUDENT(student_id),
  FOREIGN KEY (exam_id) REFERENCES EXAM(exam_id)
);

CREATE TABLE QUESTION(
  question_id INT PRIMARY KEY,
  exam_id INT,
  question_text VARCHAR(255) NOT NULL,
  marks INT,
  question_type VARCHAR(20),
  FOREIGN KEY (exam_id) REFERENCES EXAM(exam_id)
);

CREATE TABLE OPTION(
  option_id INT PRIMARY KEY,
  question_id INT,
  option_text VARCHAR(100),
  is_correct BOOLEAN,
  FOREIGN KEY (question_id) REFERENCES QUESTION(question_id)
);
```

```
CREATE TABLE ANSWER(
  answer_id INT PRIMARY KEY,
  attempt_id INT,
  question_id INT,
  option_id INT,
  written_answer VARCHAR(255),
  marks_obtained INT,
  FOREIGN KEY (attempt_id) REFERENCES ATTEMPT(attempt_id),
  FOREIGN KEY (question_id) REFERENCES QUESTION(question_id),
  FOREIGN KEY (option_id) REFERENCES OPTION(option_id)
);

CREATE TABLE RESULT(
  result_id INT PRIMARY KEY,
  attempt_id INT,
  score INT,
  grade VARCHAR(5),
  status VARCHAR(20),
  FOREIGN KEY (attempt_id) REFERENCES ATTEMPT(attempt_id)
);
```

2.4 Insertion of tuples into the table – DML commands

```
INSERT INTO STUDENT VALUES
(1,'Supriya','Yadav','2006-05-10','supriya@gmail.com','pass123','9876543210'),
(2,'Ankita','Majhi','2006-07-15','ankita@gmail.com','pass678','9876543219');
INSERT INTO EXAM VALUES
(101,'DBMS Test','2026-03-10',60);
INSERT INTO QUESTION VALUES
(1,101,'What is DBMS?',5,'MCQ'),
(2,101,'Define ER Model.',5,'DESCRIPTIVE');
INSERT INTO OPTION VALUES
(1,1,'Database Management System',1),
(2,1,'Operating System',0),
(3,1,'Compiler',0),
(4,1,'Network Device',0);
INSERT INTO ATTEMPT VALUES
(1001,1,101,1,'10:00:00','11:00:00');
INSERT INTO ANSWER VALUES
(1,1001,1,1,NULL,5);
INSERT INTO RESULT VALUES
(1,1001,10,'A','Pass');
```

OUTPUT :

For **STUDENT**

student_id	first_name	last_name	age	email_id	password	phone_no
1	Supriya	Yadav	19	supriya@gmail.com	pass123	9876543210
2	Ankita	Majhi	19	ankita@gmail.com	Pass678	9876543219

For **EXAM**

exam_id	exam_name	date	duration
101	DBMS Test	2026-03-10	60

For **OPTION**:

option_id	question_id	option_text	is_correct
1	1	Database Management System	1
2	1	Operating System	0
3	1	Compiler	0
4	1	Network Device	0

For **ANSWER**:

answer_id	attempt_id	question_id	option_id	written_answer	marks_obtained
1	1001	1	1	NULL	5
2	1001	2	NULL	ER model represents entities and relationships	5

For **QUESTION**:

question_no	exam_id	Question_text	marks	optionA	option B	option C	optionD
1	101	What is DBMS?	5	A	B	C	D
2	101	What is DBMS?	5	A	B	C	D

For **ATTEMPTS**:

attempt_no	student_id	exam_id	start_time	end_time
1001	1	101	10:00:00	11:00:00

For **RESULT**:

result_id	attempt_no	score	grade	status
1	1001	8	A	Pass